Mac Lee, PhD

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Physics PhD; applied ML; statistics & optimization; high-performance computing; author of a production open-source library (9.5M+ downloads); co-maintainer of a large-scale open-source ML framework

Education

University of California, San Diego **October 2025**
PhD in Physics, Specializing in Physical Oceanography La Jolla, CA

- Studied ocean models using analytical methods and used Python, Rust and MPI heavily in high performance simulations. Optimized numerical convergence behavior using series acceleration methods.
- Analyzed multiple terabytes of simulation data using statistical methods and regression
- Relevant courses: PHYS 239 Nonlinear Dynamics and Mathematical Modeling; CSE 230 Principles of Programming Languages; CSE 291 Program Synthesis; CSE 231 Advanced Compiler Design; CSE 291 Browser and Wasm Security

California State University, Northridge **May 2018**
M.S. in Physics with Distinction, Specializing in Condensed Matter Theory Northridge, CA

- Wrote high performance Python code for large scale numerical simulations. Optimized Hamiltonian generator function to achieve a 20x runtime reduction (reduced from 1 hour to 3 minutes per call). Optimized memory usage using quantum number conservation (over 250x reduction in memory use). Scientific computing on CPU clusters.
- Analyzed data from collaborators and performed visualization using linear regression and Fourier analysis

Experience

FastVideo **May 2026 – present**
Collaborator / Co-maintainer

- Open-source inference and post-training framework for accelerated video generation; 3.5k+ GitHub stars, adopted/forked by SGLang for multimodal inference, integrated as a backend in NVIDIA Dynamo, and used by Tencent HY-WorldPlay. https://github.com/hao-ai-lab/FastVideo
- Co-authored commits to port FLUX.2, a text-to-image flow-matching model, into FastVideo
- Ongoing work with a reinforcement learning training and inference pipeline, a performance benchmark regression tracking redesign, and Trajectory Distribution Matching, a distillation framework

University of California, San Diego **March 2022 – September 2024**
Associate / Senior TA / Lab Teaching Assistant Coordinator La Jolla, CA

- Supervised and trained lab teaching assistants and graders; held office hours; contributed to full curriculum rewrite

Projects

- Project: https://crates.io/crates/atree
  - An open-source high performance, arena-based tree structure with node removal written in Rust
  - Published on crates.io, downloaded over 9.6 million times, and used by C2PA, an Adobe backed non-profit organization.
- Project: https://github.com/macthecadillac/spike-train-autoencoder
  - Spike-Train Autoencoder - Spiking Neural Network for Image Encoding
  - A proof-of-concept image autoencoder using leaky integrate-and-fire (LIF) neuron layers, exploring how spiking neural networks can serve as biologically-grounded feature encoders
  - Encoder, decoder, and adapter trained on the STL-10 dataset with embeddings aligned to OpenAI CLIP representations

Skills

- Physical simulation; Dynamical systems; Data visualization; Fourier analysis
- 10 years of experience with Python, Rust, Numpy, SciPy, Matplotlib, git
- Other technical skills: PyTorch; Pandas; Sympy; Julia; Haskell; OCaml; TypeScript; LLVM IR; WebAssembly; MPI; Docker

Selected Publications

- M. Lee and S. G. Llewellyn Smith, “SQG point vortex dynamics with order Rossby corrections”, Eur. J. Mech. B Fluids 118, 204467 (2026).
- M. Lee and S. G. Llewellyn Smith, “Stability of SQG Kolmogorov Flow”, Dyn. Atmos. Oceans, 101680 (2026).

Awards, Grants and Honors

- 2024 Office of Naval Research Grant (University of California, San Diego)
- 2018 Mack I. Johnson Research Award for Outstanding Graduate Student in the College of Science and Mathematics (California State University, Northridge)
- 2018 C. Y. Liang Outstanding Graduate Student Award (California State University, Northridge)